# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 6 — MATHS TWO-CHAPTER TEST SET D

Chapters 5 & 6: **Prime Time, Perimeter and Area** | RBSE Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** The prime factorisation of 12 is: 1

**(a)** 3 × 4 **(b)** 2 × 3 × 3 **(c)** 2 × 2 × 3 **(d)** 2 × 6

**2.** The smallest composite number is: 1

**(a)** 3 **(b)** 6 **(c)** 4 **(d)** 2

**3.** The HCF of 15 and 25 is: 1

**(a)** 4 **(b)** 5 **(c)** 10 **(d)** 6

**4.** The HCF of 9 and 12 is: 1

**(a)** 6 **(b)** 3 **(c)** 4 **(d)** 2

**5.** The LCM of 3 and 5 is: 1

**(a)** 10 **(b)** 30 **(c)** 18 **(d)** 15

**6.** The cost of painting a floor depends on its: 1

**(a)** perimeter **(b)** angle

**(c)** number of corners **(d)** area

**7.** The area of a parallelogram-shaped figure can be found by converting it to a: 1

**(a)** rectangle **(b)** circle **(c)** line **(d)** cube

**8.** Area of a rectangle of length 12 cm and breadth 7 cm is: 1

**(a)** 19 cm² **(b)** 38 cm² **(c)** 84 cm² **(d)** 96 cm²

**9.** Area of a rectangle of length 25 cm and breadth 4 cm is: 1

**(a)** 125 cm² **(b)** 100 cm² **(c)** 29 cm² **(d)** 58 cm²

**10.** Area of a triangle with base 20 cm and height 11 cm is: 1

**(a)** 110 cm² **(b)** 220 cm² **(c)** 31 cm² **(d)** 130 cm²

## SECTION — B (Very Short Answer Type) 4 × 2 = 8

*Answer in about 10–15 words each.*

**11.** Write all the factors of 17. 2

**12.** Find the smallest number divisible by both 6 and 8. 2

**13.** State how a diagonal divides a rectangle. 2

**14.** How many 1 cm squares fit into a 6 cm by 4 cm rectangle? 2

## SECTION — C (Short Answer Type) — 4 × 3 = 12

*Answer in about 30–40 words each.*

**15.** A number is divisible by both 4 and 6. Is it divisible by 24? Explain with an example. 3

**16.** Find the number of 2 cm × 2 cm tiles to cover a 20 cm × 12 cm rectangle. 3

**17.** Why is the divisibility rule of 3 true for 123? Verify. 3

OR

What are the different kinds of numbers: prime, composite, co-prime, twin primes? Give an example each. 3

**18.** A rectangle has area 36 cm². List three possible pairs of whole-number sides and the perimeters. 3

OR

The perimeter of a triangle is 30 cm and two sides are 9 cm and 11 cm. Find the third side. 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 80–100 words each.*

**19.** (i) Why do we use prime factorisation? 5
(ii) Express 252 and 420 as prime products.
(iii) Find their HCF.

OR

(i) Show that 13 is prime. 5
(ii) Check 91 and 87 for primality.
(iii) Which numbers up to 100 can be quickly ruled out?

**20.** (i) Why is the area of a triangle with the same base and height as a rectangle half of it? 5
(ii) Compare a triangle and a rectangle with base 10 and height 6.

OR

(i) What are the units used for area and perimeter? 5
(ii) Convert 4.5 m² to cm² and 8,000 cm to m.
(iii) Give two real-life examples of each concept.

**END OF QUESTION PAPER** — **Total: 40 Marks**

MASTERMIND